OPERATION Department
SIMHADRI

Note No :1

Ref:SMTPP/OPERATION/2025-26/Oper LMI/478616 **Date:**30/11/2025(17:06:35)
Subject:LMI for Isolation for plant safety.

Respected Sir/Madam ,

The LMI for plant safety revision-2 is being put for approval. This revision has no changes over earlier revision.
It is reviewed as periodical LMI review guidline. Document reviewed on 19.07.2023 and 27.05.2024
No changes envisaged.

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Note No:5

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Unauthorized Use Prohibited

Note No:11

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Note No:12

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Note No:13

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Note No:14

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Note No:17

LMI is in line with the OGN. May please be approved.

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Note No:20

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NTPC SIMHADRI

LOCATION MANAGEMENT INSTRUCTION

Isolation for Plant Safety

LMI/OGN/GEN/002

Rev: 02 MAY-2024

Approved for Implementation by HOP (SIMHADRI)

LMI APPROVAL DOCUMENT

1	**LMI NAME**	**Isolation for Plant Safety.**
2	LMI No.	LMI/OGN/GEN/002 Rev:02 June:2021
3	Reference Document	COS-ISO-00-OGN/GEN/002/ Rev:02 June:2021
4	Superseded Document	LMI/SMSTPS/OIN/GEN/002 Rev:01 June:2019
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6	Checked By	Name: P. Naga Suresh, DGM (Operation)
7	Checked By	Name: Somnath Bhattacharya AGM (Operation)
9	Checked By	Name: B.K.Rathore, AGM (Safety)
10	Checked By	Name: Nilanjan Mandal, AGM (TMD)
11	Checked By	Name: Anil Maurya, AGM (C&I)
12	Checked By	Name: M.K.Jana, AGM (EMD)
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15	Checked By	Name: Jayamol Rachel, AGM (MTP)
16	Checked By	Name: Navin Kishore, AGM (EEMG)
17	Reviewed By	Name: Hemendra Verma G M (Maintenance)
18	Reviewed By	Name: A.K.Raza G M (O&M)
19	Approved By	Name: Sameer Sharma H O P (Simhadri)

INDEX

TITLE	Isolation for Plant Safety
LMI No.	LMI/OGN/GEN/002 Rev:02

ISOLATION FOR PLANT SAFETY

1.0 INTRODUCTION

When isolation of plant is requested or instructed for plant safety reasons it should not be affected by valves, switches or other devices capable of movement unless they have been positively Isolated/De-energized/Immobilized. Where physical disconnection is impracticable, a direct check on the isolation should be included in the pre-start and pre-commissioning check lists.

This LMI is issued to comply with Operation Guidance note OGN/GEN/002 Rev: 02. This LMI gives directives on how to isolate plant and equipment safely, and how to reduce the risk of releasing hazardous substances during intrusive activities such as
maintenance and sampling operations.

2.0 SUPERSEDED DOCUMENTS

Suspended: LMI/ OGN/GEN/002; ISSUE NO.:1 REV: 1 DATE: OCT2019

3.0 REVISION DETAILS:

6.3.5 AND 7.1 CORRECTIONS DONE.

1) IN 6.3.5 “ISOLATION” CHANGED TO “SYSTEM”.

2) IN 7.1 SENIOR OPERATION ENGINEER CHANGED TO SENIOR AUTHORIZED PERSON.

4.0 SCOPE:

This LMI is applicable to all 4x500 MW units of SIMHADRI stage1&2 and their individual/common auxiliaries have electrical and mechanical systems with high pressure & high velocity water, steam, air or gas & electricity at high/low voltage, fuel oil, Lube oil and Ash slurry system. This LMI also considers the safety of plant and covers isolations necessary where some plant feature introduces an unacceptable risk of damage. Prudence suggests that similar isolations should be effected to separate redundant plant from operational plant, even when no risk is apparent.
By extension, this LMI also considers the safety of personnel in that, damage to a plant item could result in injury to persons nearby. This LMI is related to a general requirement of Indian Factory Act - 1948 (Amendment 1987) to provide safe plant.
Further requirement of this Act is not within the scope of this LMI and is covered in NTPC Safety policy, standing instructions and Safety Rules.

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LMI No.	LMI/OGN/GEN/002 Rev:02

5.0 DISCUSSION

The salient facts giving rise to the requirement are as follows:

5.1 In certain equipment or plant isolations, valves are closed, supply is isolated and a chain lock is being applied to hand wheels.

5.2 However, the hand wheels are normally connected to the valve disc by a complex mechanism and, regardless of the locked hand wheel, the valve disc could move without giving the operators an obvious warning. Plant operating personnel to take cognizance of this factor and ensure correct choice of method of isolation

5.3 PLANT IDENTIFICATION

A scheme to uniquely identify all process plant, piping, and valves should be drawn up. All items should be readily identifiable on the plant and referenced on the piping and instrumentation diagrams (P&IDs). In addition, Formal, simple, easily visible and unambiguous labeling should be provided wherever mistakes in identification could occur and could result in significant consequences.

5.4 LOCATION OF ISOLATION AND TESTING FACILITIES

Unless risk assessment indicates otherwise, isolation and bleed points should be as close as possible to the plant item. Concentration of maintenance work in one place aids control of the isolation arrangements and minimizes the amount of water/steam/oil to be depressurized/ drained. Ensure that bleeds are:

- Arranged so that their discharge cannot harm personnel or plant and
- Hazardous material can be conveyed to a safe place for disposal; and easily accessible for checking

5.5 ACCESS AND LIGHTING

Design should enable suitable access to, and lighting of, all isolation points and associated items. Where barriers need to be inserted or where sections of plant or pipe work need to be removed, lifting beams suitable for attachment of portable lifting appliances, or access for cranes, should be provided.

5.6 HUMAN FACTORS

The performance of isolations depends not only on the integrity of the isolations done, but also on the adequacy of the arrangements for identifying each isolation point, securing the isolation, proving/monitoring and maintaining overall control of work

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5.7 TRAINING AND COMPETENCE

All personnel involved in the isolation of plant and equipment should be competent to carry out their responsibilities. They should understand the purpose, principles and practices of isolation procedures and safety rules – for their own role, and for others involved in the operation of isolations systems; and be aware of the following:

- Hazards: A detailed understanding of the potential consequences of any release of hazardous substances and hazards on the plant and from adjacent plant.
- Documentation: Understand the P&IDs, loop diagrams, cause and effect diagrams and power supplies applicable to the isolation.
- PTW (permit-to-work): Understand the system of PTW and isolation certificates in use and know the procedures for issuing PTW and for identifying what isolations are required. All relevant persons should have the knowledge of E&M Safety Rule Handbook.
- Isolation procedures: Good working knowledge of Isolation and risk assessment procedures for plant,
  - Installing/removing isolations,
  - Draining, venting, purging and flushing,
  - How to test and confirm correct installation of isolations.
  - How to record isolations on an isolation certificate and
  - Be able to assess the risks from non-standard isolations.

6.0 IMPLEMENTATION OF THE REQUIREMENT

It is recommended that isolations within the scope of this LMI should take the following forms. Methods 1 and 2 (in order of preference), should be used when the isolation is expected to be of long duration. Method 3 should only be used for isolations of short
duration, or shall be changed to method 1 or method 2 as soon as practicable.

6.1 METHOD -1

A complete disconnection should prevent flow as follows

e.g. By the removal of a section of pipe work

Responsibility: MM

e.g. Removal of an electrical conductor, or the removal of switch modules

Responsibility: EM

e.g. Racking of breaker/ module

Responsibility: Operation

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6.2 METHOD-2

A barrier should be inserted to prevent flow,

e.g. By means of a spade or blank in pipelines

Responsibility: MM

This barrier should be adequate to withstand the maximum pressure differential. If such a barrier is situated adjacent to a mechanism such as butterfly valve the barrier should be adequate to sustain, without failure, any force that the mechanism could exert on it, should that mechanism be operated inadvertently. Barriers should be as obvious as is practicable.

Responsibility: MM

6.3 METHOD-3

6.3.1 A valve should be locked in the closed position with a solid connection between the locking device and the actual pipe closure. OR a main electrical switch should be locked in the open position

Responsibility: OPN/MM/EM

6.3.2 All control, indication and auxiliary power supplies for the valve operating mechanism or electrical switch should be disconnected.

Responsibility: OPN /EM

And such disconnections should be protected by locking or by warning notices to avoid tampering. Responsibility: OPN

Here `disconnection' means electrical switches open position locked, fuses removed, Section of pipe removed, etc.

Responsibility: OPN /EM/MM

6.3.3 And All stored energy capable of operating any mechanism within the equipment Should be reduced to zero. If any mechanism within the equipment can be operated by a spring or Instrument air (or other stored energy device), then, when the plant is isolated, this spring must be released, or, Instrument air shall be disconnected. If this is impractical, the locking feature (s) must not only lock the valve disc but must also effectively restrain the spring or Pneumatic energy.
Responsibility: OPN /EM/MM/C&I

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6.3.4 All test procedures within the limits of the isolation should be suspended. Appropriate notices to this effect should be affixed in the area.

Responsibility: OPN /EM/MM/C&I

6.3.5 One permit shall not allow multiple systems. Specific permit shall be taken for every system.

Responsibility: OPN /EM/MM/C&I

7.0 INSTRUCTIONS

7.1 The change-over from method 3, either to method 1 or to method 2, should be carried out either with the plant shut down completely, or under the supervision of a Senior Authorized Person who is fully aware of the history and the risks to the plant involved.
Responsibility: SCE

7.2 The choice of method to be adopted, in a particular case, should be approved by the station head of O&M or Senior personnel authorized by him to make the choice.
Compliance with standing orders should prevent unauthorized alternations being made to such isolations and should ensure that all persons likely to be concerned are aware of the situation.

8.0 RESPONSIBILITY:

HOD(O) is responsible for implementation of this LMI

9.0 REFERENCE DOCUMENT

COS-ISO-00-OGN/GEN/002, Rev.No :2 June 2021, OPERATION GUIDANCE NOTE on “Isolation for Plant Safety”

10.0 REVIEW:

Head (O) shall arrange for review and take the approval of competent authority. The document will be reviewed 2 yearly basis or as and when required.